

Program Question_7

```
Implicit none
Integer,Parameter::max_n=50
Integer::n
Real(8),dimension(0:max_n)::a,exact
Real(8)::Errorr

!Initial Condition
a(0)=1.0
a(1)=5.0
! calculate Recurrence Value
Do n=2,max_n
    a(n)=2.0*a(n-1)-a(n-2)+3.0

End Do
Write(*,*)"n      |Recurrence      |Exact_value      |Error"
Write(*,*)"-----"
Do n=0,max_n
    exact(n)=1.5*n**2+2.5*n+1.0
    Errorr=abs(a(n)-exact(n))
    Write(*,'(I3,3X,F12.2,F10.2,F12.2)')n,a(n),exact(n),Errorr
End Do
```

End Program